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Assignment-2 (Class-Work-Questions)

C Programming — Topic: Decision Making

Question On IF

Q1. Positive Number

Problem: Input a number and print "Positive" if the number is greater than 0.

Code:

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num > 0)
    {
        printf("Number is a Positive number\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter a number: 50
Number is a positive number.
```

Q2. Even Number

Problem: Input a number and print "Even" if the number is divisible by 2.

Code:

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 2 == 0)
    {
```

```
        printf("Number is an even number\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter a number: 44
Number is an even number.
```

Q3. Divisible by 5

Problem: Input a number and print "Divisible by 5" if it is divisible by 5.

Code:

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 5 == 0)
    {
        printf("Number is divisible by 5.\n");
    }

    return 0;
}
```

Output:

```
Enter a number: 55
Number is divisible by 5.
```

Same idea as the even number check, but this time we divide by 5 instead of 2. Any number whose remainder with 5 is zero is a multiple of 5.

Q4. Positive and Even

Problem: Check whether a number is positive and even using if.

Code:

```
#include <stdio.h>

int main()
{
    int num;
```

```
printf("Enter a number: ");
scanf("%d", &num);

if (num > 0 && num % 2 == 0)
{
    printf("Number is positive and even.\n");
}

return 0;
}
```

Output:

```
Enter a number: 50
Number is positive and even.
```

Q5. Salary Check

Problem: Input salary and print "High Salary" if salary is greater than ₹50,000.

Code:

```
#include <stdio.h>

int main()
{
    float salary;

    printf("Enter your number: ");
    scanf("%f", &salary);

    if (salary > 50000)
    {
        printf("High Salary\n");
    }

    return 0;
}
```

Output:

```
Enter a number: 600000
High Salary
```

Questions on if-else

Q1. Positive or Negative

Problem: Input a number and check whether it is positive or negative.

Code:

```
#include <stdio.h>

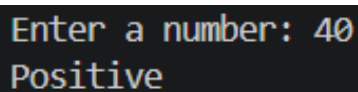
int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num >= 0)
    {
        printf("Positive\n");
    }
    else
    {
        printf("Negative\n");
    }

    return 0;
}
```

Output:



```
Enter a number: 40
Positive
```

Q2. Pass or Fail

Problem: Input marks and print "Pass" if marks are 40 or above, "Fail" otherwise.

Code:

```
#include <stdio.h>

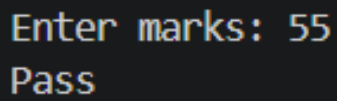
int main()
{
    int marks;

    printf("Enter marks: ");
    scanf("%d", &marks);

    if (marks >= 40)
    {
        printf("Pass\n");
    }
    else
```

```
    {  
        printf("Fail\n");  
    }  
  
    return 0;  
}
```

Output:



```
Enter marks: 55  
Pass
```

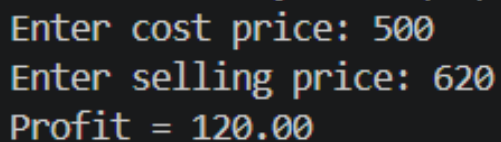
Q3. Profit or Loss

Problem: Input cost price and selling price. Print whether there is a profit or loss.

Code:

```
#include <stdio.h>  
  
int main()  
{  
    float cp, sp;  
  
    printf("Enter cost price: ");  
    scanf("%f", &cp);  
  
    printf("Enter selling price: ");  
    scanf("%f", &sp);  
  
    if (sp > cp)  
    {  
        printf("Profit = %.2f\n", sp - cp);  
    }  
    else if (sp < cp)  
    {  
        printf("Loss = %.2f\n", cp - sp);  
    }  
    else  
    {  
        printf("No profit, no loss\n");  
    }  
  
    return 0;  
}
```

Output:



```
Enter cost price: 500  
Enter selling price: 620  
Profit = 120.00
```

Q4. Number is Zero or Not

Problem: Check whether a number is zero or non-zero.

Code:

```
#include <stdio.h>

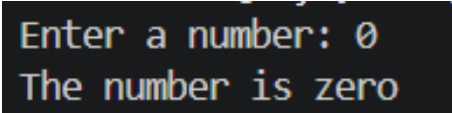
int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num == 0)
    {
        printf("The number is zero\n");
    }
    else
    {
        printf("The number is non-zero\n");
    }

    return 0;
}
```

Output:



```
Enter a number: 0
The number is zero
```

Q5. Divisible by 3

Problem: Check whether a number is divisible by 3 or not.

Code:

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 3 == 0)
    {
        printf("Divisible by 3\n");
    }
    else
    {
        printf("Not divisible by 3\n");
    }
}
```

```
}  
  
return 0;  
}
```

:

```
Enter a number: 36  
Divisible by 3
```

Questions on else-if Ladder

Q1. Grade Calculation

Problem: Input marks and display grade: 90-100 -> A, 80-89 -> B, 70-79 -> C, 60-69 -> D, below 60 -> F.

Code:

```
#include <stdio.h>

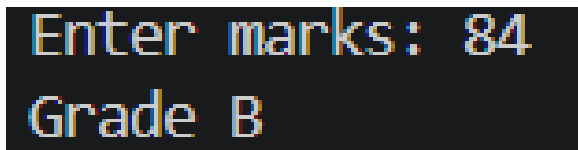
int main()
{
    int marks;

    printf("Enter marks: ");
    scanf("%d", &marks);

    if (marks >= 90 && marks <= 100)
    {
        printf("Grade A\n");
    }
    else if (marks >= 80)
    {
        printf("Grade B\n");
    }
    else if (marks >= 70)
    {
        printf("Grade C\n");
    }
    else if (marks >= 60)
    {
        printf("Grade D\n");
    }
    else
    {
        printf("Grade F\n");
    }

    return 0;
}
```

Output:

A screenshot of a terminal window with a black background. The text 'Enter marks: 84' is displayed on the first line, and 'Grade B' is displayed on the second line. Both lines of text have a rainbow-colored outline effect.

Q2. Number Range

Problem: Check whether a number is: 1-10 -> Small, 11-50 -> Medium, 51-100 -> Large, above 100 -> Very Large.

Code:

```
#include <stdio.h>
```

```

int main()
{
    int num;

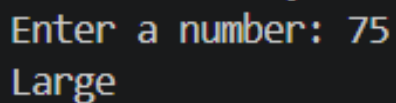
    printf("Enter a number: ");
    scanf("%d", &num);

    if (num >= 1 && num <= 10)
    {
        printf("Small\n");
    }
    else if (num <= 50)
    {
        printf("Medium\n");
    }
    else if (num <= 100)
    {
        printf("Large\n");
    }
    else
    {
        printf("Very Large\n");
    }

    return 0;
}

```

Output:



```

Enter a number: 75
Large

```

Q3. Temperature

Problem: Input temperature and display: above 40 -> Very Hot, 30-40 -> Hot, 20-29 -> Normal, 10-19 -> Cold, below 10 -> Very Cold.

Code:

```

#include <stdio.h>

int main()
{
    float temp;

    printf("Enter temperature (in C): ");
    scanf("%f", &temp);

    if (temp > 40)
    {
        printf("Very Hot\n");
    }
}

```

```

    else if (temp >= 30)
    {
        printf("Hot\n");
    }
    else if (temp >= 20)
    {
        printf("Normal\n");
    }
    else if (temp >= 10)
    {
        printf("Cold\n");
    }
    else
    {
        printf("Very Cold\n");
    }

    return 0;
}

```

Output:

```

Enter temperature (in C): 33
Hot

```

Q4. Age Category

Problem: Input age and display: 0-12 -> Child, 13-19 -> Teenager, 20-59 -> Adult, 60+ -> Senior Citizen.

Code:

```

#include <stdio.h>

int main()
{
    int age;

    printf("Enter your age: ");
    scanf("%d", &age);

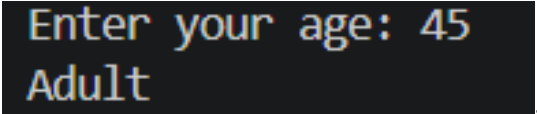
    if (age >= 0 && age <= 12)
    {
        printf("Child\n");
    }
    else if (age <= 19)
    {
        printf("Teenager\n");
    }
    else if (age <= 59)
    {
        printf("Adult\n");
    }
    else

```

```
{
    printf("Senior Citizen\n");
}

return 0;
}
```

Output:



Enter your age: 45
Adult

Q5. Electricity Bill

Problem: Input units and calculate the category: 0-100 -> Low usage, 101-200 -> Medium usage, 201-300 -> High usage, above 300 -> Very high usage.

Code:

```
#include <stdio.h>

int main()
{
    int units;

    printf("Enter units consumed: ");
    scanf("%d", &units);

    if (units <= 100)
    {
        printf("Low usage\n");
    }
    else if (units <= 200)
    {
        printf("Medium usage\n");
    }
    else if (units <= 300)
    {

```

```
        printf("High usage\n");
    }
    else
    {
        printf("Very high usage\n");
    }

    return 0;
}
```

Output:

```
Enter units consumed: 250
High usage
```

Questions on Nested if

Q1. Positive Even/Odd

Problem: Using nested if, check whether a number is: Positive Even, Positive Odd, or Negative.

Code:

```
#include <stdio.h>

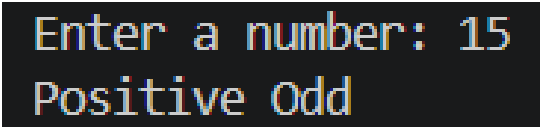
int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num >= 0)
    {
        if (num % 2 == 0)
        {
            printf("Positive Even\n");
        }
        else
        {
            printf("Positive Odd\n");
        }
    }
    else
    {
        printf("Negative\n");
    }

    return 0;
}
```

Output:



```
Enter a number: 15
Positive Odd
```

Q2. Largest of Three Numbers

Problem: Using nested if, find the largest among three numbers.

Code:

```
#include <stdio.h>

int main()
{
    int a, b, c;
```

```

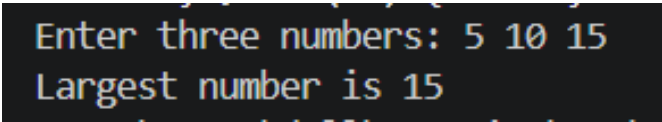
printf("Enter three numbers: ");
scanf("%d %d %d", &a, &b, &c);

if (a >= b)
{
    if (a >= c)
    {
        printf("Largest number is %d\n", a);
    }
    else
    {
        printf("Largest number is %d\n", c);
    }
}
else
{
    if (b >= c)
    {
        printf("Largest number is %d\n", b);
    }
    else
    {
        printf("Largest number is %d\n", c);
    }
}

return 0;
}

```

Output:



```

Enter three numbers: 5 10 15
Largest number is 15

```

Q3. Student Eligibility

Problem: Input attendance percentage and marks. A student is eligible for the exam only if attendance is at least 75%. If eligible, check whether the student has passed based on marks.

Code:

```

#include <stdio.h>

int main()
{
    float attendance;
    int marks;

    printf("Enter attendance percentage: ");
    scanf("%f", &attendance);

    if (attendance >= 75)
    {

```

```
printf("Enter marks: ");  
scanf("%d", &marks);  
  
if (marks >= 40)  
{  
    printf("Eligible and Passed\n");  
}  
else  
{  
    printf("Eligible but Failed\n");  
}  
}  
else  
{  
    printf("Not Eligible for the exam\n");  
}  
  
return 0;  
}
```

Output:

```
Enter attendance percentage: 80  
Enter marks: 55  
Eligible and Passed
```

Questions on switch-case

Q1. Day of the Week

Problem: Using switch-case, input a number from 1 to 7 and display the corresponding day.

Code:

```
#include <stdio.h>

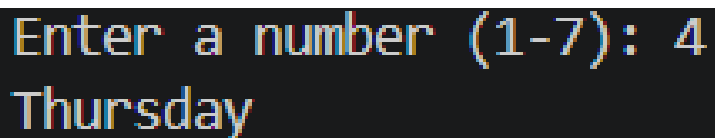
int main()
{
    int day;

    printf("Enter a number (1-7): ");
    scanf("%d", &day);

    switch (day)
    {
        case 1:
            printf("Monday\n");
            break;
        case 2:
            printf("Tuesday\n");
            break;
        case 3:
            printf("Wednesday\n");
            break;
        case 4:
            printf("Thursday\n");
            break;
        case 5:
            printf("Friday\n");
            break;
        case 6:
            printf("Saturday\n");
            break;
        case 7:
            printf("Sunday\n");
            break;
        default:
            printf("Invalid input\n");
    }

    return 0;
}
```

Output:



```
Enter a number (1-7): 4
Thursday
```

Q2. Vowel or Consonant

Problem: Using switch-case, input a character and check whether it is a vowel (a, e, i, o, u) or not.

Code:

```
#include <stdio.h>

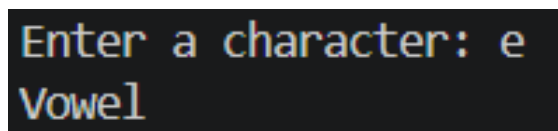
int main()
{
    char ch;

    printf("Enter a character: ");
    scanf("%c", &ch);

    switch (ch)
    {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
        case 'A':
        case 'E':
        case 'I':
        case 'O':
        case 'U':
            printf("Vowel\n");
            break;
        default:
            printf("Consonant\n");
    }

    return 0;
}
```

Output:



```
Enter a character: e
Vowel
```

Q3. Number to Word

Problem: Using switch-case, input a number from 0 to 9 and display it in words. Example: Input 5, Output Five.

Code:

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number (0-9): ");
```

```

scanf("%d", &num);

switch (num)
{
    case 0:
        printf("Zero\n");
        break;
    case 1:
        printf("One\n");
        break;
    case 2:
        printf("Two\n");
        break;
    case 3:
        printf("Three\n");
        break;
    case 4:
        printf("Four\n");
        break;
    case 5:
        printf("Five\n");
        break;
    case 6:
        printf("Six\n");
        break;
    case 7:
        printf("Seven\n");
        break;
    case 8:
        printf("Eight\n");
        break;
    case 9:
        printf("Nine\n");
        break;
    default:
        printf("Invalid number\n");
}

return 0;
}

```

Output:

```

Enter a number (0-9): 5
Five

```

Q4. Traffic Signal

Problem: Using switch-case, input a character: R -> Red -> Stop, Y -> Yellow -> Wait, G -> Green -> Go.

Code:

```
#include <stdio.h>
```

```

int main()
{
    char signal;

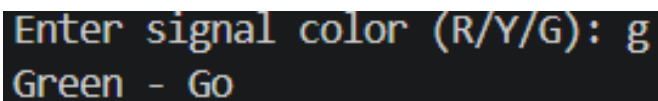
    printf("Enter signal color (R/Y/G): ");
    scanf("%c", &signal);

    switch (signal)
    {
        case 'R':
        case 'r':
            printf("Red - Stop\n");
            break;
        case 'Y':
        case 'y':
            printf("Yellow - Wait\n");
            break;
        case 'G':
        case 'g':
            printf("Green - Go\n");
            break;
        default:
            printf("Invalid signal\n");
    }

    return 0;
}

```

Output:



```

Enter signal color (R/Y/G): g
Green - Go

```

Q5. Grade Description

Problem: Input a grade character (A, B, C, D, F) and display: A -> Excellent, B -> Very Good, C -> Good, D -> Average, F -> Fail.

Code:

```

#include <stdio.h>

int main()
{
    char grade;

    printf("Enter grade (A/B/C/D/F): ");
    scanf("%c", &grade);

    switch (grade)
    {
        case 'A':

```

```

        printf("Excellent\n");
        break;
    case 'B':
        printf("Very Good\n");
        break;
    case 'C':
        printf("Good\n");
        break;
    case 'D':
        printf("Average\n");
        break;
    case 'F':
        printf("Fail\n");
        break;
    default:
        printf("Invalid grade\n");
}

return 0;
}

```

Output:

```

Enter grade (A/B/C/D/F): B
Very Good

```

Q6. Calculator with Nested Switch

Problem: Using nested switch-case, build a calculator with two categories - 1. Arithmetic (Addition, Subtraction, Multiplication, Division) and 2. Relational (Greater, Smaller).

Code:

```

#include <stdio.h>

int main()
{
    int category, choice;
    float a, b;

    printf("1. Arithmetic\n");
    printf("2. Relational\n");
    printf("Choose a category: ");
    scanf("%d", &category);

    printf("Enter two numbers: ");
    scanf("%f %f", &a, &b);

    switch (category)
    {
        case 1:
            printf("1. Addition\n");
            printf("2. Subtraction\n");

```

```

printf("3. Multiplication\n");
printf("4. Division\n");
printf("Choose operation: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        printf("Result = %.2f\n", a + b);
        break;
    case 2:
        printf("Result = %.2f\n", a - b);
        break;
    case 3:
        printf("Result = %.2f\n", a * b);
        break;
    case 4:
        if (b != 0)
        {
            printf("Result = %.2f\n", a / b);
        }
        else
        {
            printf("Division by zero is not possible\n");
        }
        break;
    default:
        printf("Invalid operation\n");
}
break;

```

```

case 2:
    printf("1. Greater\n");
    printf("2. Smaller\n");
    printf("Choose operation: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            if (a > b)
                printf("%.2f is greater\n", a);
            else
                printf("%.2f is greater\n", b);
            break;
        case 2:
            if (a < b)
                printf("%.2f is smaller\n", a);
            else
                printf("%.2f is smaller\n", b);
            break;
        default:
            printf("Invalid operation\n");
    }

```

```
        }  
        break;  
  
        default:  
            printf("Invalid category\n");  
    }  
  
    return 0;  
}
```

Output:

```
1. Arithmetic  
2. Relational  
Choose a category: 1  
Enter two numbers: 10 50  
1. Addition  
2. Subtraction  
3. Multiplication  
4. Division  
Choose operation: 3  
Result = 500.00
```

```
1. Arithmetic  
2. Relational  
Choose a category: 2  
Enter two numbers: 50  
40  
1. Greater  
2. Smaller  
Choose operation: 2  
40.00 is smaller
```